Compound Net

The Combination Network and Concatenation
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1. Purpose

This document describes how concatenation and the combination network work in Compound Net. It is the part of the architecture where the locked professors' outputs, the original input and the domain fingerprint (Di) are brought together into a final answer. The document complements the scientific final report and the note on fingerprint handling.

2. The Role of Concatenation

In the locked architecture the professors receive only the original input X. They are specialists in their domains and each deliver a scalar judgement. The combination network, by contrast, receives three sources of information at the same time:

- X – the 12 original data points
- The professor outputs – three scalars (out_PFE, out_PFF, out_PFK)
- Di – the fingerprint indicating domain and difficulty

These three are joined into a single vector. That is the concatenation. The combination network sees everything at once and can learn for itself how to weigh the information.

3. What the Concatenation Looks Like

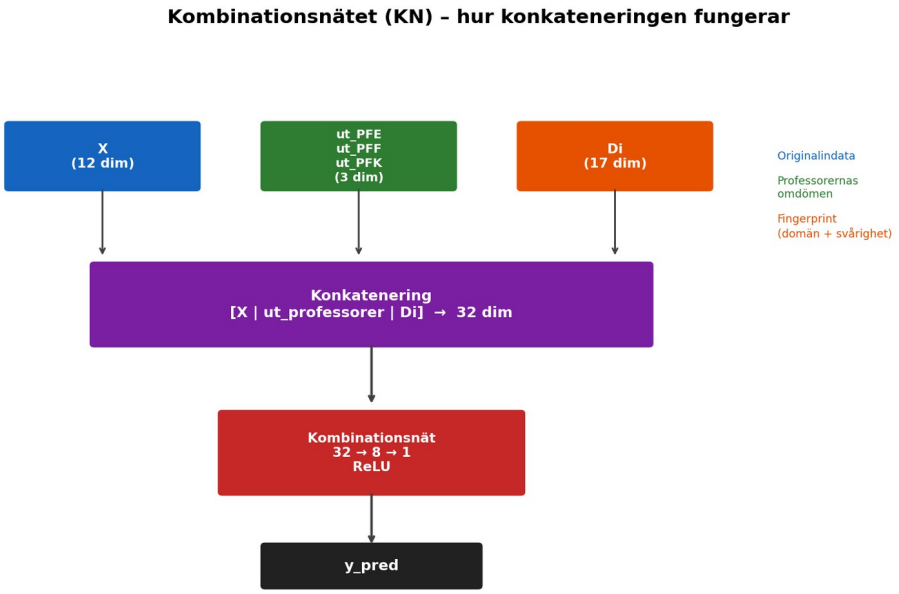


Figure 1. Concatenation into the combination network.

The resulting input vector to the combination network has 32 dimensions when Di is 17-dimensional:

$[x_1 \ldots x_{12} \mid out_PFE \mid out_PFF \mid out_PFK \mid d_1 \ldots d_{17}]$

4. Configuration of the Combination Network

In the experiments reported here the following configuration was used:

Property	Value
Input	32 dim (12+3+17)
Architecture	32 → 8 → 1
Activation	ReLU (hidden), linear (output)
Optimisation	AdamW, lr=0.001, wd=0.01
Degree of freedom	≥ 10:1

5. Results

With comparable professors (PFE, adjusted PFF, PFK) and Di 17-dim concatenation the combination network reached the following result on mixed data:

Component	Relative RMSE
PFE	30.7%
PFF (adjusted)	39.3%
PFK	16.4%
Combination network	**16.7%**

This is a significant improvement compared with earlier runs without Di or with uneven professors, where the result was around 80–85% or 30–33%.

6. What Concatenation Enables

By seeing X, the professor outputs and Di at the same time the combination network can:

- Trust a professor more when Di indicates that it is the right domain
- Rely more on the original data when the professors are uncertain
- Use the difficulty level in Di to adjust its confidence

It is this weighing together that allows a small combination network to reach lower error than any single professor on mixed data.

7. Conclusion

Concatenation is the mechanism that makes Compound Net more than the sum of its professors. By giving the combination network both the experts' judgements and a balanced fingerprint the system can handle mixed data better than any single specialist. The result of 16.7% relative RMSE with comparable professors and Di 17-dim shows that the method works.